

M. Ryleigh Davis

Planetary Science Ph.D. Candidate

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RyleighDavis

Education

Ph.D.	California Institute of Technology , Planetary Science	Sept 2020 to May 2026
M.Sc.	California Institute of Technology , Planetary Science	May 2022
M.Sc.	Northern Arizona University , Applied Physics	Aug 2018 to May 2020
	• Thesis: Low Energy Electron Sputtering of Water Ice	
B.Sc.	University of Arizona , Astronomy & Physics, minor in Mathematics	Aug 2013 to May 2017

Work & Research Positions

Graduate Student Researcher , Caltech - Division of Geological and Planetary Sciences	CA, USA
• Lead multi-wavelength (UV—mid-IR) observational programs using HST, JWST, and ground-based facilities to characterize the surface composition of icy satellites. Developed mineralogical interpretation frameworks integrating laboratory spectra and spectral modeling. Advisor: Mike Brown	Aug 2020 to current
Senior Associate Consultant , Clarity Insights: Data Analytics Consulting	IL, USA
• Lead the team developing a platform for distributed computations on the cloud (via pySpark and AWS) for a major financial institution and worked with the Data Science team to implement machine learning and predictive algorithm tools for data analytics.	July 2017 to Aug 2018 1 year 1 month
Telescope Operator , University of Arizona - Steward Observatory	AZ, USA
• Operated telescopes for research data acquisition and trained/certified new observers (Bok 2.2m/B&C Spec, Bok 2.2m/90Prime, Kuiper 1.55m/Mont4k, Ray White 21").	Jan 2014 to May 2017 3 years 5 months
• Assisted Mountain Operations with Instrument changes, system testing, cleaning of mirrors and filters and other telescope maintenance projects as a summer intern.	
NASA Space Grant , University of Arizona, Research Intern	AZ, USA
• Worked with Dr. Caitlin Griffith to observe and reduce Kuiper 1.55m telescope data to examine and characterize the atmospheres of hot Jupiter exoplanets	Aug 2015 to May 2016 1 year

Selected Awards

- **Graduate Fellow**, Caltech Center for Comparative Planetary Evolution (3CPE) (2021-current)
- **Alexander F. H. Goetz Fellowship**, Caltech (2020-2021)
- **Grad College/GSG Academic Scholarship Award**, Northern Arizona University (2019)
- **Undergraduate Research Achievement**, University of Arizona Department of Astronomy (2017)
- **Arizona Excellence Award**, University of Arizona (2013-2017)

Invited Seminars & Colloquia

- **Carnegie EPL**, Astronomy Seminar, Mar. 27, 2026 (upcoming)
- **University of Arizona**, Origins Seminar Series, Mar. 9, 2026 (upcoming)
- **UCSD/SDSU**, Astronomy & Astrophysics Colloquium, Mar. 11, 2026 (upcoming)
- **UCLA**, Astronomy Lunch Seminar, Dec. 9, 2025
- **Uranian Rings & Small Satellites Working Group**, Science Talk, Dec. 5, 2025
- **JPL**, Icy Worlds Collaboration and Exchange (ICE) Seminar, Apr. 10, 2025
- **Europa Clipper Team**, Composition Working Group Science Talk, Jan. 23, 2024
- **Caltech**, DIX Planetary Science Seminar, Jun. 7, 2022; May 9, 2023; Jun. 4, 2024; Apr. 15, 2025

Publications

First-Author

- **Davis, M.R.**, Trumbo, S.K., Brown, M.E, and Belyakov, M. (2025). “Spectroscopic Mapping of Callisto with HST/STIS and Implications for its Surface Composition”. The Planetary Science Journal, 6, 161.
- **Davis, M.R.** and Brown, M.E. (2024). “Pwyll and Manannán Craters as a Laboratory for Constraining Irradiation Timescales on Europa”. The Planetary Science Journal, 5, 107.
- **Davis, M.R.**, Brown, M.E., & Trumbo, S.K. (2023). “The Spatial Distribution of the Unidentified 2.07 μm Absorption Feature on Europa and Implications for its Origin”. The Planetary Science Journal, 4(8), 148.
- **Davis, M.R.**, Meier, R.M., Cooper, J.F., and Loeffler, M.J. (2021). “The contribution of electrons to the sputter-produced O₂ exosphere on Europa”. The Astrophysical Journal Letters, 908(2), L53.
- **Fitzpatrick (Davis), M.R.** (2020). “Low Energy Electron Sputtering of Water Ice” (M.Sc. Thesis, Northern Arizona University).

Submitted/In Review

- **Davis, M.R.**, Belyakov, M., Wong, I., Milby, Z., and Brown, M.E (2026). “Triton’s Wake: Neptune’s Shattered Inner Moons Reveal Icy Body Interior Compositions”. Science Advances (in review).

Co-Authored

- Carmack, R.A., **Davis, M.R.**, and Loeffler, M. (2026). “Electron-induced sputtering from damaged H₂O-ice”. (in prep).
- Belyakov, M., **Davis, M.R.**, Wong, I., Batygin, K., and Brown, M.E. (2026). “Nereid as a Regular Satellite of Neptune”. Science Advances (in review).
- Belyakov, M., Wong, I., Bolin, B.T., **Davis, M.R.**, Bromley, S.J., Lisse, C.M., Brown, M.E. (2026). “The Volatile Inventory of 31/ATLAS as seen with JWST/MIRI”. ApJL (in review)
- Brown, M.E, Belyakov, M., Chandra, S., **Davis, M.R.**, McDowell, M., Pandya, A., Trinh, K., and Trumbo, S.K. (2026). “Such stuff as moons are made on: Deuterated water and the formation of the satellites of Uranus”. PNAS (in review).
- Brown, M.E, Trumbo, S.K., **Davis, M.R.**, and Chandra, S. (2025). “Deuterated water on the satellites of Saturn”. The Planetary Science Journal, 6, 229.
- Brown, M.E, Trumbo, S.K., Belyakov, M., **Davis, M.R.**, and Pandya, A. (2025). “A JWST study of CO₂ on the satellites of Saturn”. The Planetary Science Journal, 6, 180.
- Belyakov, M. et al. (inc. **Davis, M.R.**) (2025). “Palomar and Apache Point Spectrophotometry of Interstellar Comet 31/ATLAS”, Research Notes of the AAS, 9, 194.
- Belyakov, M., **Davis, M.R.**, Milby, Z., Wong, I., and Brown, M.E. (2024). “JWST Spectrophotometry of the Small Satellites of Uranus and Neptune”. The Planetary Science Journal, 5(5), 119.
- Kunimoto, M., Vanderburg, A., Huang, C.X., **Davis, M.R.**, et al. (2023). “TOI-4010: A System of Three Large Short-period Planets with a Massive Long-period Companion”. The Astronomical Journal, 166(1), 7.
- Trumbo, S.K., **Davis, M.R.**, Cassese, B., and Brown, M. E. (2022). “Spectroscopic Mapping of Io’s Surface with HST/STIS: SO₂ Frost, Sulfur Allotropes, and Large-scale Compositional Patterns”, The Planetary Science Journal, 3(12), 272.
- Zellem, R.T. et al. (inc. **Fitzpatrick (Davis), M.R.**)(2015). “XO-2b: A hot jupiter with a variable host star that potentially affects its measured transit depth”, The Astrophysical Journal, 810(1), 11.

Grants and Observing Programs

Principal Investigator (PI) Programs:

- Tracing the Origin of Hydrated Minerals on the Small Satellites of Uranus and Neptune (PI); Palomar/NGPS (2025B); 5 nights
- Constraining the Composition and Thermal Histories of Silicate Minerals on Callisto (PI); JWST/MIRI Cycle 3 GO (2024-2025); 15.96 hrs
- Reconstructing the Histories of the Ice Giant Systems through Small Satellite Observations (Co-PI); JWST/NIRSpec Cycle 3 GO (2024-2025); 26.71 hrs
- Photometric U and B Band Observations of Bright Transiting Exoplanets to Constrain their Atmospheric Compositions (Undergraduate PI); Kuiper/Mont4K & Bok/90Prime (2014-2017); 37 nights

Co-Investigator (Co-I) Programs:

- MIRI MRS Observations of the Third Interstellar Object (PI: Belyakov); JWST/MIRI Cycle 4 DDT (2025); 9.2 hrs
- The Aftermath of a Transient Deposition Event on Europa (PI: Trumbo); JWST/NIRSpec Cycle 3 DDT (2024); 13.2 hrs
- The Saturnian satellites as a laboratory for CO₂ in the outer solar system (PI: Brown); JWST/NIRSpec Cycle 2 GO (2023-2024); 21.2 hrs
- Complete NIRSpec coverage of Europa's surface: CO₂, salt hydrates, and the potential for unexpected discovery (PI: Trumbo); JWST/NIRSpec Cycle 2 GO (2023-2024); 21.2 hrs
- Titan's Tropospheric Methane Abundance Across the Seasons (PI: Griffith); NASA Solar System Workings (2019-2022)
- A Global Map of Titan's Tropospheric Methane Abundance Near Northern Solstice (PI: Griffith); Keck II/NIRSPEC (2017-2019); 10 half nights

Teaching Positions

Graduate Teaching Assistant, California Institute of Technology

AZ, USA

- Teaching Assistant for Ge 108: Applications of Physics to the Earth Sciences (3 terms)

Oct 2021 to Dec 2023

Graduate Teaching Assistant, Northern Arizona University

AZ, USA

- Instructor for Introductory Electricity and Magnetism laboratory class (4 semesters).
- Teaching Assistant for Indigenous Astronomy (2 semesters).

Aug 2018 to May 2020

Student Educator and Video Producer, Steward Observatory Education Group

AZ, USA

- Communicated with and educated the public on Astronomy and science related topics via production of YouTube videos on the Active Galactic Videos channel.
- Aided in curriculum development, filming, and video editing for an Introductory Astronomy online course (hosted on Coursera) with Professor Chris Impey.

Aug 2015 to May 2017
1 year 9 months

Special Projects Assistant for Education & Public Outreach, National Optical Astronomy Observatory (NOAO)

AZ, USA

- Developed Astronomy and STEM related education curriculum and outreach activities, coordinated and ran public events, managed the Globe at Night citizen science campaign, and ran teacher training's and workshops on Astronomy and Space Education.
- Co-ran an outreach and education training session at the AAS Winter Meeting.

Aug 2014 to Oct 2015
1 year 2 months

Engineering & Robotics Instructor, Engineering for Kids Summer Camp

AZ, USA

- Taught science, engineering, and robotics summer camp curriculum including Lego Robotics for students aged 3-7 and Engineering Basics and Design for students 8-14.

May to Aug 2014
3 months

Science Communication & Outreach

- **University of Arizona College of Science Career Center**; Panelist for Earth, Physical, & Space Science Graduate Student Panel (March 19, 2025)
- **Caltech GPS Buddy Program**; Mentor (2021-2025)
- **Caltech GPS Division Planetary Science Option Ombuds Person** (2021-2022)
- **Caltech Women Mentoring Women**; Mentor (2021-2022)
- **Skype a Scientist**; Scientist Partner (2020-current)
- **Letters to a Pre-Scientist**; Scientist/STEM Professional (2022-2024)
- **Society of Women in Space Exploration (SWISE)**; NAU chapter co-founder & Vice President (2018-2020)
- **Middle School Science Mentor**; Sinagua Middle School (Flagstaff, AZ) (2019-2020)
- **Flagstaff Community Star Party**; Public lecture (2019)
- **University of Arizona Astronomy Club**; President & Outreach Coordinator (2013-2017)

Selected Conference Talks

- **Davis, R.** & Brown, M. (Nov 2025) "From UV to Mid-IR: Illuminating the Nature of Callisto's Dark Material", Jovian Icy Moons' Surface Interactions With Their Environment Workshop (Madrid, Spain)
- **Davis, R.**, Belyakov, M., Brown, M., and Wong, I. (Sept 2025) "JWST Reveals Phyllosilicates on the Small Inner Moons of Neptune", EPSC-DPS2025 #181 (Helsinki, Finland)
- **Davis, M.R.**, Belyakov, M., Wong, I., and Brown, M.E. (Jun 2025) "JWST Spectroscopy of the Small Inner Satellites and Rings of Uranus and Neptune", OPAG Meeting 2025 #6008 (Tucson, AZ)
- **Davis, M.R.** and Brown, M.E. (Jan. 2025) "Exploring the Composition of Callisto's Dark Material with JWST MIRI MRS", JWST Solar System Science Workshop (Meudon, France)
- **Davis, M.R.**, Trumbo, S.K., and Brown, M.E. (Dec 2024) "Spectroscopic Mapping of Callisto's Surface with HST/STIS", AGU Fall Meeting P32A-06 (Washington D.C.)
- **Davis, M.R.** and Brown, M.E. (Dec 2023). "Pwyll and Manannán Craters as a Laboratory for Constraining Irradiation Timescales on Europa", AGU Fall Meeting P43B-04 (San Francisco, CA).
- **Davis, M.R.** and Brown, M.E. (Oct 2023). "Pwyll Crater as a Laboratory for Understanding Irradiation Timescales on Europa", Workshop on the Origins and Habitability of the Galilean Satellites (Marseille, France)
- **Davis, M.R.** and Brown, M.E. (Oct 2022). "A Critical Analysis of the 2.07 μm Absorption Feature on Europa", DPS Meeting #54 (London, Canada).
- **Fitzpatrick (Davis), M.R.** and Griffith, C.A. (Jan 2020). "New Reduction Techniques for Echelle Spectroscopy of Extended Objects: Creating a Global Map of Titan's Tropospheric Methane Humidity", AAS Meeting #235 (Honolulu, HI).
- **Fitzpatrick (Davis), M. R.** and Loeffler, M. J. (Dec 2019). "The Sputtering Yield of Electron Irradiated Water Ice and Its Dependence on Temperature", AGU Fall Meeting (San Francisco, CA).
- **Fitzpatrick (Davis), M. R.** and Loeffler, M. J. (Sept 2019). "Composition of the Sputtered Flux in Electron Irradiated Ices", Flagstaff Astronomy Symposium (Flagstaff, AZ).
- **Fitzpatrick (Davis), M.R.**, Martins-Filho, W., Griffith, C.A., Pearson, K., Zellem, R.T., and AzGOE (Oct. 2016). "A Study of the Effects of Underlying Assumptions in the Reduction of Multi-Object Photometry of Transiting Exoplanets", DPS Meeting #48 (Pasadena, CA).
- **Fitzpatrick (Davis), M.R.**, Watson, Z., Zellem, R.T., Pearson, K., Griffith, C.A., and AzGOE (Jan 2015). "High-precision ground-based observations of transiting exoplanets to detect their magnetic fields and undiscovered companions", AAS Meeting #225 (Seattle, WA).